Exam3 Practice Problems

1. Find the curl and divergence. Use curl to determine whether the vector field is conservative.
   (a) $\mathbf{F} = x^2y\,\mathbf{i} - (z^3 - 3x)\,\mathbf{j} + 4y^2\,\mathbf{k}$
   (b) $\mathbf{F} = (3x + 2z^2)\,\mathbf{i} + (\frac{x^3y^2}{z})\,\mathbf{j} - (z - 7x)\,\mathbf{k}$
   (c) $\mathbf{F} = 6x\,\mathbf{i} + (2y - y^2)\,\mathbf{j} + (6z - x^3)\,\mathbf{k}$
   (d) $\mathbf{F} = (z\cos y)\,\mathbf{i} - (xz\sin y)\,\mathbf{j} + (x\cos y)\,\mathbf{k}$
2. Determine whether or not the vector $\mathbf{F}(x, y, z) = (2xy^2 + z)\mathbf{i} + (2yx^2)\mathbf{j} + x\mathbf{k}$ is conservative. If it is, compute the potential function.
3. Determine whether or not the vector $\mathbf{F}(x, y, z) = (yz + e^x)\mathbf{i} + (xz + e^y)\mathbf{j} + (xy + e^z)\mathbf{k}$ is conservative. If it is, compute the potential function.
4. Evaluate $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$ and $\mathbf{F} = (yz^2)\mathbf{i}+(xz^2)\mathbf{j}+(2xyz)\mathbf{k}$, where C is a path from $(0, 0, 0)$ to $(1, 1, 1)$.
5. Evaluate $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$ and $\mathbf{F} = (yz + e^x)\mathbf{i} + (xz + e^y)\mathbf{j} + (xy + e^z)\mathbf{k}$, where C is a path from $(0, 0, 0)$ to $(1, 1, 1)$.
6. Let $\mathbf{F} = y^2\mathbf{i} - x\mathbf{j}$, and let C be a triangle with vertices $(0, 0), (1, 0), (1, 1)$ traversed counterclockwise. Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
7. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
8. Let $\mathbf{F} = (y + x)\mathbf{i} + (y - x)\mathbf{j}$, and let C be a circle with $x^2 + y^2 = 4$ traversed counterclockwise. Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
9. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
10. Let $\mathbf{F} = (5x)\mathbf{i} + (x + 3y)\mathbf{j}$, and let C be the square obtained by going from $(0, 0)$ to $(1, 0)$, from $(1, 0)$ to $(1, 1)$, from $(1, 1)$ to $(0, 1)$, and from $(0, 1)$ to $(0, 0)$ (traveling along a straight horizontal or vertical line each time). Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
11. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
12. Let $\mathbf{F} = (x^2)\mathbf{i} + (yx)\mathbf{j}$, and let C be curve obtained by going from $(0, 0)$ to $(1, 1)$ along the graph of the curve $y = x^2$, and then from $(1, 1)$ to $(0, 0)$ along the graph of $y = x$. Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
13. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
14. Consider the surface $\mathbf{r}(u, v) = \langle u + v, u^2 - v^2, uv\rangle$, for $0 \le u \le 1$, $0 \le v \le 2$. Set up, but do not solve, an integral which represents the area of this surface.
15. Consider the surface $\mathbf{r}(u, v) = \langle u^2v, uv^2, u + v\rangle$, for $0 \le u \le 1$, $0 \le v \le 1$. Suppose the density of this surface is given by the function $\delta(u, v) = e^{u+v}$. Set up, but do not solve, an integral which represents the mass of this surface.

16. Consider the surface obtained by revolving the graph of the function $y = e^{5x}$, for $0 \leq x \leq 1$, around the x-axis. Set up, but do not solve, an integral which represents the area of this surface.

17. Consider the surface obtained by revolving the graph of the function $y = 1/\sqrt{x}$, for $1 \leq x \leq 4$, around the y-axis. Let $f(x, y) = y^5$. Set up, but do not evaluate, the surface integral $\int\int_S f \, d\sigma$.

18. Consider the surface consisting of the portion of the cylinder $x^2 + y^2 = 1$ which is above $z = 0$ and below $z = 1$. Let $f(x, y, z) = x^2 z^2$. Evaluate the surface integral $\int\int_S f \, d\sigma$.

19. Consider the surface consisting of the portion of the cone $z = \sqrt{x^2 + y^2}$ which is above the plane $z = 2$ and below the plane $z = 4$. Let $f(x, y, z) = z^2$. Evaluate the surface integral $\int\int_S f \, d\sigma$

20. Consider the surface consisting of the portion of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 1$. Find the area of this surface.

21. Consider the surface consisting of the portion of the sphere $x^2 + y^2 + z^2 = 1$ which is below the upward facing cone $z = \sqrt{x^2 + y^2}$ and above the downward facing cone $z = -\sqrt{x^2 + y^2}$. Find the area of this surface.

22. Consider the surface consisting of the portion of the plane $2x + 3y + z = 6$ which is in the first octant. Find the area of this surface.

23. Let S be the outward oriented sphere $x^2 + y^2 + z^2 = 1$, and let $\mathbf{F} = xy\mathbf{i} + 2\mathbf{j} - z\mathbf{k}$ be a vector field. Set up, but do NOT solve, the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

24. Let S be the surface consisting of the portion of the cone $z = \sqrt{x^2 + y^2}$ which is above the plane $z = 2$ and below the plane $z = 4$, oriented so that the normal is always pointing down. Let $\mathbf{F} = y\mathbf{i} + 1\mathbf{k}$ be a vector field. Compute the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, dS$.

25. Let S be the surface given by the portion of the plane $x + 3y + 3z = 9$ which lies in the first octant, oriented so that the normal always points in the positive z direction. Let $\mathbf{F} = \langle 3, 1, -2 \rangle$ be a vector field. Compute the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

26. Let S be the surface consisting of the portion of the plane $3x + 2y + 5z = 30$ which lies in the first octant, oriented upwards. Let $\mathbf{F} = \langle y, x, x \rangle$ be a vector field. Compute the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

27. Consider the surface consisting of the portion of the graph of $z = x^2 y^2 + x + y$ which is above the square $0 \leq x \leq 1$, $0 \leq y \leq 1$, oriented up. Let $\mathbf{F}(x, y) = 1\mathbf{i} + 2\mathbf{j} + xy^2\mathbf{k}$. Set up, but do not evaluate, the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

28. Let S be the surface given by the portion of the graph of $z = x^4 + y^2$ which is above the square $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, oriented up. Let $\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + z^2\mathbf{k}$ be a vector field. Set up, but do NOT solve, the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

29. Let S be the outward oriented top half of the sphere $x^2 + y^2 + z^2 = 1$ for $z > 0$. Let $\mathbf{F} = \langle y, 0, -z^2 \rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$.

30. Let S be the downward oriented surface consisting of the portion of the sphere $x^2 + y^2 + z^2 = 4$ which lies below the plane $z = -1$. Let $\mathbf{F} = \langle 1 - z, -3y, x \rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$.

31. Let S be the outward oriented surface given by the portion of the cylinder $x^2+y^2=1$ which is above the plane $z=-1$ and below the plane $z=1$. Let $\mathbf{F}=\langle z, z^2, z^3\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

32. Let S be the inward surface given by the portion of the cylinder $x^2+y^2=1$ which is above the plane $z=0$ and below the plane $z=5$. Let $\mathbf{F}=\langle yz, 0, 2z\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

33. Let S be the downward oriented surface consisting of the portion of the cone $z=\sqrt{x^2+y^2}$ which is above the plane $z=1$ and below $z=3$. Let $\mathbf{F}=\langle 1-z, -3y, x\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

34. Let S be the outward oriented surface consisting of the top hemisphere $x^2+y^2+z^2=25$ and the disk area $x^2+y^2\leq 25$ at height $z=0$. Let $\mathbf{F}=\langle x^2y, x^3, yx^2\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

35. Let S be the outward oriented cube with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, $(1,1,0)$, $(1,0,1)$, $(0,1,1)$ and $(1,1,1)$. Let $\mathbf{F}=\langle y^2, xy^2, xz^2\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

36. Find a parameterization for each surface:

    (a) The portion of the surface $z=10$ that is above the square $-1\leq x\leq 1$, $-2\leq y\leq 2$.

    (b) The paraboloid $z=x^2+y^2$ that is above the triangle with vertices at (0,0), (0,1) and (1,1).

    (c) The paraboloid $z=x^2+y^2$ that is above the circular disk of radius 3 centered at the origin.

    (d) The portion of the cone $z=\sqrt{x^2+y^2}$ that lies between the cylinders $x^2+y^2=4$, $x^2+y^2=16$.

    (e) The portion of the plane $x-y+z=4$ that is within the cylinder $x^2+y^2=9$.

    (f) The portion of cylinder $x^2+y^2=1$ that lies between the planes $z=0$ and $z=3$ in the first octant.

    (g) The portion of cylinder $x^2+y^2=4$ that lies between the plane $z=y-2$ and $z=x+4$.

    (h) The portion of the sphere $x^2+y^2+z^2=4$ that lies inside the cylinder $x^2+y^2=2$ and above the xy-plane.

37. Let f be a scalar function defined on $\mathbb{R}^3$ and let $\mathbf{F}$ be a vector field defined on $\mathbb{R}^3$. Assume that f and all components of vector field $\mathbf{F}$ have continuous second partial derivatives. Select all expressions that are well-defined.

    (a) $\nabla\cdot(\nabla\times\mathbf{F})$

    (b) $\nabla\times(\nabla\cdot\mathbf{F})$

    (c) $\nabla(\nabla\times\mathbf{F})$

    (d) $\nabla\times(\nabla\times\mathbf{F})$

    (e) $(\nabla f)\cdot(\nabla\times\mathbf{F})$

    (f) $\nabla\cdot(\nabla\cdot\mathbf{F})$

    (g) None of the above are correct

38. Select all true statements about curl.

    (a) $\operatorname{curl}\mathbf{F}$ at a point measures the tendency of the field to rotate around that point.
    (b) The $\operatorname{curl}\mathbf{F}$ points along the axis of rotation.
    (c) If $\operatorname{curl}\mathbf{F} = 0$ everywhere, the field $\mathbf{F}$ is constant.
    (d) For $\mathbf{F} = \langle -y, x, 0\rangle$, $\operatorname{curl}\mathbf{F} = 2\mathbf{k}$.

39. Select all true statements about divergence.

    (a) $\operatorname{div}\mathbf{F} > 0$ at a point means that point acts like a source.
    (b) If $\operatorname{div}\mathbf{F} = 0$ everywhere, the field is incompressible.
    (c) If $\operatorname{div}\mathbf{F} = 0$ everywhere, then $\mathbf{F}$ must also have $\operatorname{curl}\mathbf{F} = \mathbf{0}$.
    (d) For $\mathbf{F} = \langle x, y, z\rangle$, $\operatorname{div}\mathbf{F} = 3$ at every point.

40. Determine whether the following statements are true or false.

    (a) The fundamental theorem of line integrals can only be applied to closed curves.
    (b) The fundamental theorem of line integrals can only be applied to conservative vector fields.
    (c) Let C be a simple closed curve in the xy-plane, oriented counterclockwise, that encloses a region R, and let $\mathbf{F} = <P, Q>$ where P and Q have continuous partial derivatives on R. Then $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$ can be evaluated using Green's Theorem, whether $\mathbf{F}$ is conservative or not.
    (d) If C is a closed curve, then $\int_C \mathbf{F}\cdot\mathbf{T}\ \ ds = 0$ for any vector field $\mathbf{F}$.
    (e) If $\mathbf{F}$ is a vector field, then $div\,\mathbf{F}$ is a vector field.
    (f) If $\mathbf{F}$ is a vector field, then $curl\,\mathbf{F}$ is a vector field.

41. Consider the following methods

    (a) Direct parameterization
    (b) Fundamental Theorem of Line Integrals
    (c) Green's Theorem
    (d) Stokes' Theorem
    (e) None of the methods without more information

    Select all methods that could be used to evaluate

    (1) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle x^2 - y,\ x + y^2\rangle$ and C is the unit circle in the xy-plane (counterclockwise).
    (2) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle 2xy,\ x^2\rangle$ and C is the boundary of the square $0 \le x \le 1$, $0 \le y \le 1$, traversed counterclockwise.
    (3) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle z,\ x,\ y\rangle$ and C is the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = y + 2$, traversed counterclockwise when viewed from above.
    (4) $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle 2xy + z,\ x^2,\ x\rangle$ and C is the curve $\mathbf{r}(t) = \langle t, t^2, t^3\rangle$ for $0 \le t \le 1$ (from $(0,0,0)$ to $(1,1,1)$).
    (5) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle y,\ z,\ x\rangle$ and C is the triangle with vertices $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, traversed counterclockwise when viewed from above.
    (6) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle 3x^2y,\ x^3 + 2y\rangle$ and C is the circle $x^2 + y^2 = 4$ in the xy-plane, traversed counterclockwise.